

Jo-Hao Huang (Andrew)

Mobile: +1 984 379 8914 | E-mail: jhandrewhuang@gmail.com | Github: <https://github.com/andrew403> |
LinkedIn: <https://www.linkedin.com/in/andrew-huang-322b97244/>

PROFESSIONAL SUMMARY

Computer Science Master's student specializing in **computer vision, 3D reconstruction, and robotics**, with hands-on experience building production-oriented perception systems across research and industry. Skilled in Python, C++, PyTorch, OpenCV, ROS2, and 3D Gaussian Splatting pipelines, with a track record of improving reconstruction quality, automating visual evaluation workflows, and deploying scalable computer vision tools. Proven ability to own complex R&D projects from prototype to integration in fast-paced, cross-functional environments.

EDUCATION

North Carolina State University (NCSU), Raleigh, NC | Dec. 2025 | **Master of Computer Science**

Relevant Coursework: Software Engineering, Machine Learning, Computer Vision, Robotics

Yuan Ze University (YZU), Taoyuan City, Taiwan | Jun 2022 | **Bachelor of Science in Computer Science and Engineering**

SKILLS

Coding Languages: C, C++, Python, Java, Rust

Technical Skills: PyTorch, OpenCV, Jetson, ROS2, COLMAP, LiDAR, IMU, Nerfstudio, 3DGS, FastAPI, Docker, Git, ONNX, TensorRT, Isaac Sim, Gazebo, CARLA, YOLO, SAM, ARKit, ZeroMQ, Wandb

Concepts: Computer Vision, Machine Learning, Deep Learning, Quantization, 3D Gaussian Splatting, SLAM, Structure-from-Motion, COLMAP, Neural Rendering, Image Segmentation, Object Detection, Depth Estimation, Pose Estimation, Sensor Fusion, Robotics Perception

WORK EXPERIENCE

Computer Vision R&D Engineer, **forREAL Inc.**, Danvers, MA | Apr. 2026 – Jul. 2026.

- Developed VLM based automatic quality scoring for COLMAP pose reconstruction and Nerfstudio/Lichtfeld Studio 3DGS outputs, replacing manual review with model-based reprocessing triggers.
- Built prompt-driven object counting for 3D Gaussian Splatting by combining segmentation masks with COLMAP/3DGS reprojection, achieving over **95%** accuracy and surpassing existing state-of-the-art methods.
- Integrated Lichtfeld Studio into the 3DGS pipeline through FastAPI endpoints and RANSAC-based ground alignment, improving reconstruction quality by up to **2x** without manual orientation correction.
- Restored metric scale from ARKit depth and enhanced iOS capture with dual-exposure SAM-based fusion, enabling spatial measurement and improving window-scene reconstruction quality by **1.5x**.

Research Assistant, **North Carolina State University**, Raleigh, NC | Apr. 2025 – Mar. 2026

- Conduct research on perception models under multi-robot scenarios in Dr. Peng Gao's lab.

Autonomous Driving Development Intern, **Industrial Technology Research Institute**, Hsinchu City | Apr. 2024 – Nov. 2024

- Reviewed computer vision research papers, deploy and conducting initial tests for the open-source models on Nvidia Orin platform.
- Implemented and fine-tuned monocular vision models for a **Level 2 Autonomous Bus**, adapting open-source architectures to perform real-time object detection, depth estimation and BEV generation within the ROS2 ecosystem.
- Modeled autonomous driving scenarios using Isaac Sim and integrated them with ROS to create **hardware-in-loop** simulation.

IT Infrastructure Intern, **Advanced Semiconductor Engineering Inc.**, Taoyuan City | Feb. 2021 – Apr. 2022

- Maintained enterprise IT infrastructure and automated software installation/update workflows for production-line and office systems.

RESEARCH & PROJECT EXPERIENCE

CERPE: Deep Multi-Robot Cooperative Localization | Second Author | **IROS2026** | [arxiv 2607.14539](#)

- Co-developed a novel localization framework for multi-robot systems, addressing challenges of limited communication bandwidth and intermittent visual overlap in ephemeral collaborative perception without relying on global references.
- Integrated VGGT model to jointly estimate ego-motion and inter-robot relative pose, and utilized Metric3Dv2 to align predicted monocular depth with metric depth references, successfully resolving monocular scale ambiguity.
- Engineered a communication architecture that decouples overlap detection from pose estimation using SALAD-based descriptors, and implemented metrically scaled ego-motion to continuously propagate and maintain relative poses during non-overlapping periods.
- Achieved up to a **72%** reduction in position error and a **90%** reduction in rotation error compared to state-of-the-art methods. Validated robust real-time performance across CARLA simulations, the OpenMars dataset, and physical Agilex Limo Pro robot teams.

Maze Solving Robot | Jan 2025 – May 2025

- Utilized Q-learning algorithm to train a robot for maze navigation via a wall-following strategy within the Gazebo simulation environment.
- Implemented the resulting trained policy on a physical Triton robot, using LiDAR sensor data for real-time obstacle avoidance.

SFML Platform Video Game | Sept. 2023 – Dec. 2023

- Created a 2D online platform game through SFML and C++, with multi-threading technique, pause/un-pause, and ZeroMQ networking features that supports multiplayer.

Promotional Robot with Human Target Recognition through Camera and Smart Insoles | Sept. 2021 – Jul. 2023

- Identified stationary characters through OpenPose with insole pressure sensor; applied it to robots and combined with user interface to achieve product promotion function.
- Ensured software run properly through manual testing of compatibility between the Nvidia TX2 Jetpack SDK version, system environment, and other packages (e.g., NumPy, Pandas).
- Summarized scattered data captured by OpenPose using Python; synchronized the time axis with insole data; organized and wrote results into multiple .csv files.